

Research paper

Physical system modeling and optimized control strategy of 6-DOF vessel motion simulator based on MBD and LBM

Weibin Ma, Peng Wang^{*}, Huachao Dong, Xinjing Wang, Hengzhi Fang, Xianxian Fan

School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an, China

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ABSTRACT

Under the wave-coupling effects, vessels experience six-degree-of-freedom (6-DOF) complex and extreme motions. Vessel motion simulators replicate wave-induced vessel motions for dynamic inspection of shipboard equipment, which requires integrated models, input diversity, and motion control accuracy. This paper establishes an integrated physical system model for the vessel motion simulator with a Stewart mechanism based on kinematic and dynamic analyses. It integrates a multi-body dynamics (MBD) model, an input and inverse kinematic system model, a sensor system model, and a control system model. The Lattice-Boltzmann method (LBM) is used for computational fluid dynamics (CFD) simulation of a vessel in waves, and 6-DOF vessel motions are predicted for the simulator inputs under multiple sea states. Furthermore, a fuzzy PID controller with fuzzy rules optimized by the genetic algorithm (GA) is proposed and its control performance, including the motion accuracy, robustness, and stationarity, is evaluated. The simulation and test analyses show that the physical system model can accurately simulate vessel motions and GA-fuzzy PID controller improves the control performance.

1. Introduction

With higher depths and larger areas of marine operations, vessels experience six-degree-of-freedom (6-DOF) complex and extreme motions comprising surge, sway, heave, roll, pitch and yaw under the wave-coupling effects (Chen et al., 2023b), as shown in Fig. 1. Violent motions of up to 40° have been reported for an APL container sailing against the waves in the North Pacific Ocean. Such extreme waves significantly impact the normal operation of vessels and shipboard equipment, as manifested by a reduction in reliability and induced damage, such as in the Suwa-Maru incident (Liu et al., 2022). To avoid such effects, it is necessary to conduct dynamic verification tests under various sea states in a realistic and safe environment for shipboard equipment (for example, the wave compensation system in Fig. 1). However, it is costly and laborious to conduct equipment tests under the dynamic response of a real target vessel in marine environments or a vessel model in large pools. With recent developments in digital simulation technology, 6-DOF vessel motion simulators have been developed, which enable maritime quality inspections using laboratory predictive evaluations (Terrero-Gonzalez et al., 2024).

Vessel motion simulators leverage the Stewart parallel mechanism to replicate vessel movement in response to waves in a mechanical manner.

It enables shipboard equipment testing in a controlled environment and effective response to diverse waves in practical applications (Gu and Song, 2017). Additionally, simulators facilitate pre-service training for practitioners and deck dynamic testing for important equipment. However, current standards of simulators lack explicit technical criteria for the Stewart mechanism to simulate vessel movement under varied sea states. To achieve realistic simulator behavior, establishing physical system models corresponding to the physical entity can aid in analyzing the overall performance in external environments.

A fully physical system model for simulators requires the application of multi-body dynamics (MBD) to describe the intricate dynamic characteristics and mutual feedback in various operations. Several studies have demonstrated the efficacy of separate MBD kinematics and dynamics, control analysis of the Stewart platform in Simulink (Furqan et al., 2017; Yin et al., 2022), MBD computer modeling (Zhang et al., 2023), and diverse design phases (Liu et al., 2023) of spatial 6-DOF parallel robots using SimMechanics. However, physical system modeling and simulation in ocean engineering field are emerging fields, and most studies focus on the modeling and simulation of a single system (Jang and Bewley, 2023; Li et al., 2012). Comprehensive integration of multiple systems for overall performance improvement and evaluation has not been achieved (Li et al., 2021). As a result, higher demands are

^{*} Corresponding author.

E-mail address: wangpeng305@nwpu.edu.cn (P. Wang).

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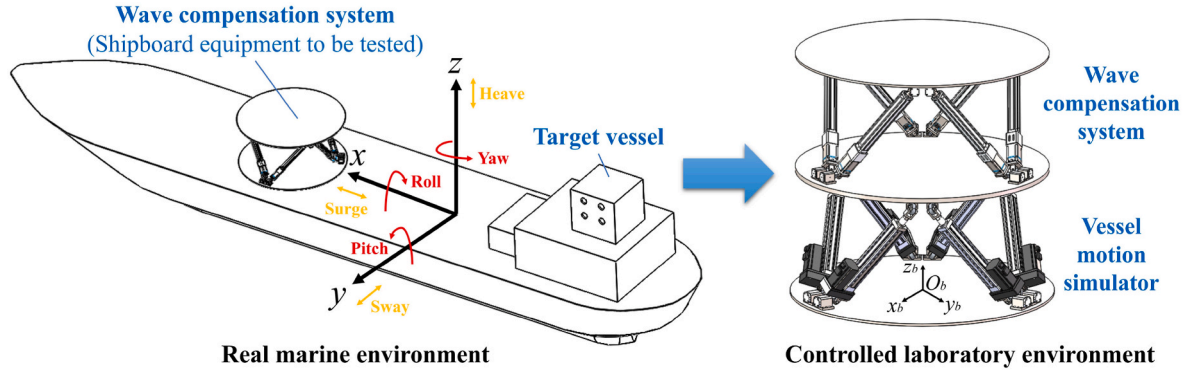


Fig. 1. 6-DOF vessel motions.

imposed on integrated models, input diversity, and full-dimensional motion accuracy of control strategies.

The randomness and variability inherent in wave and vessel motions introduce uncertainties to the input diversity of simulators. Therefore, accurately predicting and inputting the extreme behavior of a vessel directly affect the quality of simulators (Donatini et al., 2025). In more complex sea states, replacing the actual motion with simplified curves and wave spectra leads to substantial discrepancies, and the experimental data are limited and homogeneous. With the development of computational fluid dynamics (CFD) techniques, practical viscous calculations can be obtained, and have been used to predict the 6-DOF behavior of vessels (He et al., 2022). Wang et al. (2022) and Liu et al. (2018) conducted CFD numerical simulations on the motion response of standard KVLCC2 ships and DTC ships. Huang et al. (2021) and Kim et al. (2017) investigated the effects on vessel motion under different wave conditions such as unidirectional and bidirectional regular waves and wave steepness. Their findings establish that CFD simulations can serve as feasible alternatives for experimental analysis of vessel behavior (Ni et al., 2020).

However, traditional CFD simulations have significant limitations in considering mesh deformation in the context of intricate models and wide-range movements (Ganesan et al., 2024). The Lattice-Boltzmann method (LBM), which has the advantage of obtaining mesh-free solutions to vessel motions in waves, has not been studied extensively in this regard. In addition, simulators focus on the entire motion process from the beginning to stability, and using CFD results as diverse inputs to simulators and physical system models is deficient (Silva et al., 2022). To enhance the realism of simulators, the study uses LBM to numerically predict 6-DOF vessel motion under various sea states during the entire

dynamic process.

A primary function of simulators is to reproduce vessel motions with excellent accuracy, which necessitates suitable control strategies for physical system models. However, it is challenging to design control strategies. Vessels are exposed to strongly random and variable external forces, stemming from dynamic sea waves. Correspondingly, systems under high load lead to inertia, randomness, and severe response hysteresis (Cai et al., 2021). Methods such as adaptive control (Chen et al., 2024) and robust control (Ono et al., 2019) based on modern control theory require precise mathematical object models, including platforms, limb actuators, loads and environmental disturbances. When models are not sufficiently accurate or dynamic characteristics are neglected, the controller performance deteriorates significantly (Min et al., 2019). Consequently, it is difficult for these controllers to fully adapt to the Stewart mechanism with a complex dynamic model, parameter perturbations and external disturbances. In light of the superiority of fuzzy control, fuzzy PID logic algorithms are widely adopted in designing motion controllers for the Stewart simulator (Qiong et al., 2013; Zhao et al., 2024). Several studies have investigated optimization methods such as particle swarm optimization algorithm and genetic algorithm (GA) for fuzzy rules and membership functions to improve the performance (Bingul and Karahan, 2022; Dai et al., 2021).

In this study, an integrated physical system model with diverse CFD simulation inputs and an optimized control strategy is proposed for complex and high-precision performance of the vessel motion simulator. Fig. 2 shows the entire technical flow chart. A fully physical system model based on kinematic and dynamic analyses is established, including an MBD model, an input and inverse kinematic system model, a sensor system model, and a control system model. Then, the 6-DOF

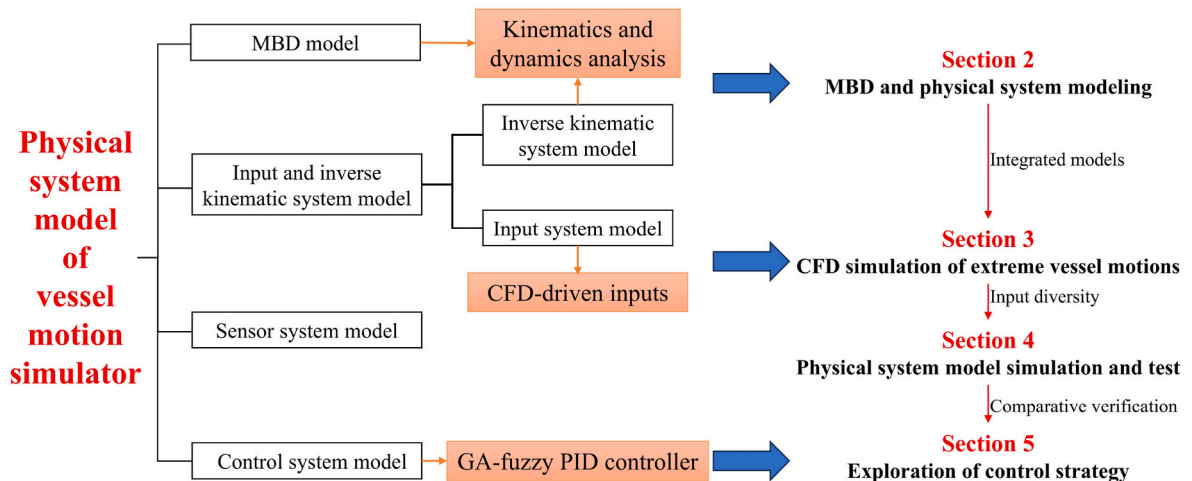


Fig. 2. Technical flow chart.

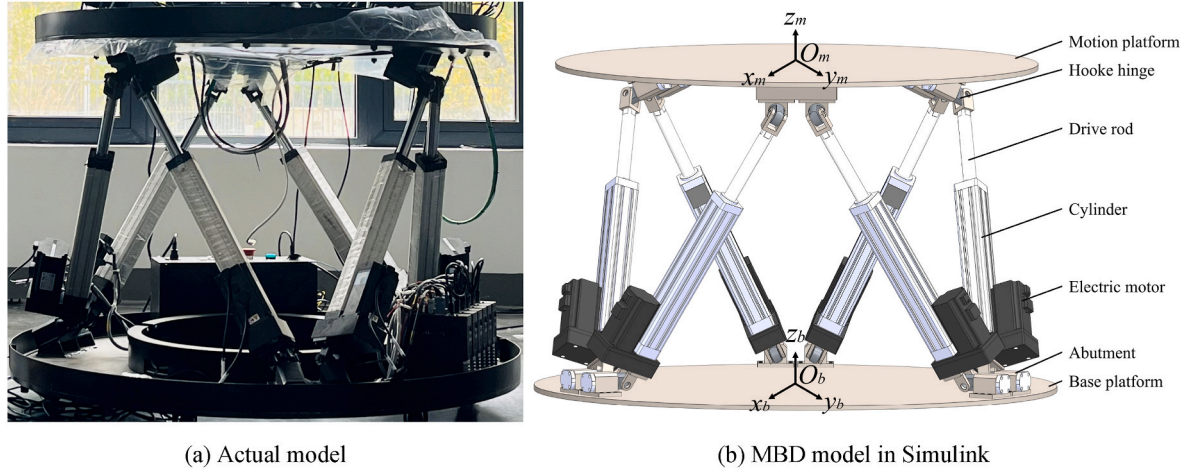


Fig. 3. Vessel motion simulator.

behavior of a vessel in waves under sea state levels 3 to 5 is obtained to develop the input system model through CFD simulations with LBM. Furthermore, a comparative analysis of the physical system model simulation and test with inputs from CFD results is conducted. Finally, a GA-fuzzy PID controller with optimized fuzzy rules is proposed for motion control, and its control performance is analyzed through the simulations.

2. Physical system modeling of the 6-DOF vessel motion simulator

To improve the performance of vessel motion simulators, it is crucial to describe the kinematic and dynamic characteristics of the system based on MBD and to adopt the integrated physical systems modeling.

2.1. Vessel motion simulator with Stewart structure

This study proposes a vessel motion simulator with the Stewart structure, as shown in Fig. 3(a). The 6-DOF vessel motion simulator is assembled by modular design, allowing to reproduce the complex behavior of vessels in a real marine environment through the MBD model. The MBD model of the actual system with a Stewart parallel mechanism is established in the form of 6-UPU, as shown in Fig. 3(b). The base and motion platforms are connected by six limb actuators and Hooke hinges with two rotational degrees of freedom. The telescopic displacement of each rod is independent, and the pose in the 6-DOF direction of the motion platform is accomplished by the combined motion of the six limb actuators.

The system is developed to input the extreme behavior of a vessel under multiple sea states and to precisely control the telescopic length of the six rods to manipulate the motion platform, including translation and rotation along the X, Y and Z axes (Lu et al., 2020). The behavior of a vessel subjected to wave disturbance is thus simulated, realizing comprehensive testing of shipboard equipment within a controlled laboratory environment.

2.2. Kinematic and dynamic analyses

To ensure that the motion platform simulates the desired 6-DOF behavior, the telescopic displacement of the six rods should be solved within the feasible range. The analysis of kinematic conditions and dynamic characteristics for the MBD model and system workspace provides key theoretical support for the physical system model and subsequent precise control.

The Stewart mechanism is a nonlinear coupled mechanism with multiple inputs and outputs. Its forward kinematic represents 18 simultaneous nonlinear equations in the six unknowns, and the inverse kinematic represents the pose of the motion platform, which is relatively easy to solve. Consequently, a mathematical model for the inverse kinematic solution is established to convert the input parameters of the motion platform into the displacement parameters of the six limbs.

To describe the movement of the mechanism, a global reference coordinate system $\{O_b\}$ and a local coordinate system $\{O_m\}$ are established at the center of the base and motion platforms, respectively. The motion acquired at the vessel mass center is converted into the inputs corresponding to the motion platform. Then, the transformations of the

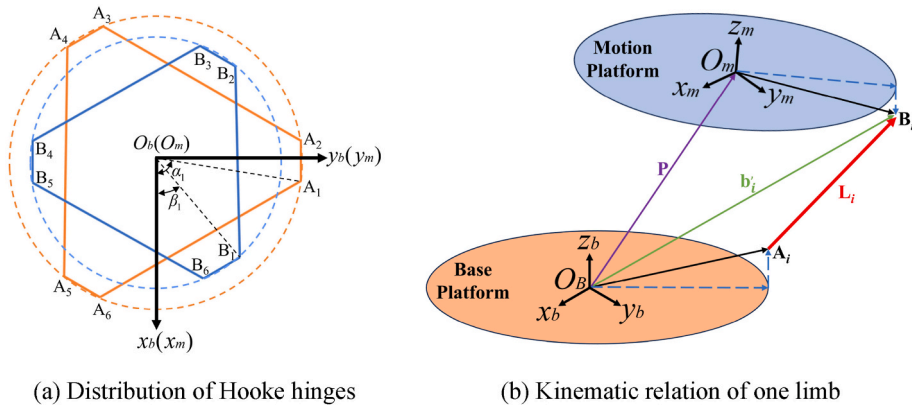


Fig. 4. Systematic inverse kinematic analysis.

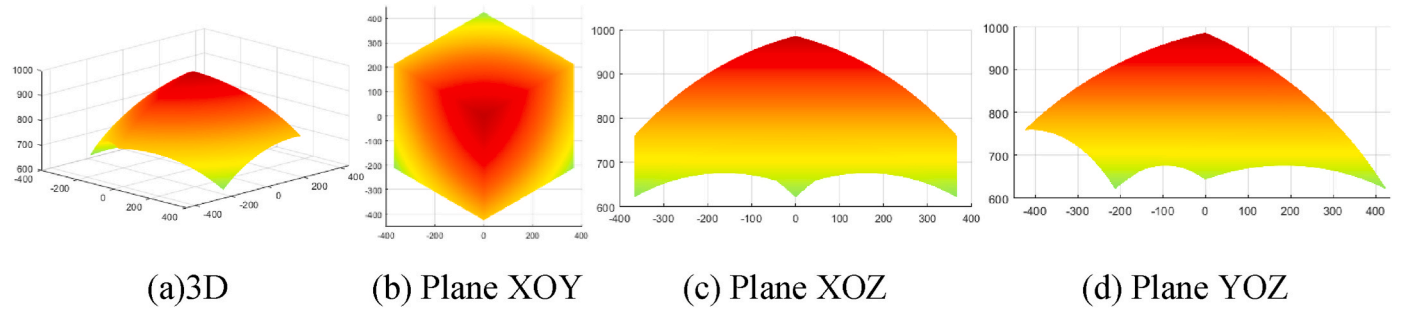


Fig. 5. System workspace.

Table 1
6-DOF operational constraints of the simulator.

Direction	Surge (X)	Sway (Y)	Heave (Z)	Roll (RX)	Pitch (RY)	Yaw (RZ)
Range	±366 mm	±422 mm	621–985 mm	±27°	±23°	±30°

translation $\mathbf{P}$ and Euler angle rotation $\mathbf{R}$ between the two coordinate systems are obtained.

Since six kinematic chains of the Stewart mechanism are symmetrical to each other, the mathematical model considers one kinematic chain as a representative example. For the i -th limb, based on the distribution of Hooke hinges shown in Fig. 4(a), the upper hinge coordinates $\mathbf{B}_i$ on the local coordinate system are transformed to the global reference coordinate system. According to Fig. 4(b) to further represent its kinematic vector relation, the inverse kinematic equation of the system can be obtained as follows:

$$\mathbf{L}_i = \mathbf{R} \cdot \mathbf{B}_i - \mathbf{A}_i + \mathbf{P} \quad (1)$$

where $\mathbf{L}_i$ is the spatial length, $\mathbf{P}$ and $\mathbf{R}$ are the linear displacement and rotation matrix of two coordinate systems, respectively, and $\mathbf{A}_i$ and $\mathbf{B}_i$ are coordinate vectors of the lower and upper hinges, respectively.

Using the pose information of two platforms, the changes in the 6-DOF position and attitude of the motion platform is converted into functions of the displacement of the six limb actuators:

$$\Delta L_i = \left| \mathbf{R} \cdot \begin{bmatrix} R_b \cdot \cos \beta_i \\ R_b \cdot \sin \beta_i \\ h_b \end{bmatrix} - \begin{bmatrix} R_a \cdot \cos \alpha_i \\ R_a \cdot \sin \alpha_i \\ h_a \end{bmatrix} + \mathbf{P} \right| - l_0 \quad (2)$$

Here, ΔL_i is the telescopic displacement of the rods, R_a and R_b are the diameter of the distribution circle of the upper and lower hinges, respectively, α_i and β_i are the angle between the upper and lower hinges and the positive direction of the x-axis, respectively, and l_0 is the initial length of limbs.

The workspace is defined as the extent to which the system can replicate the extreme position and attitude conditions. In this study, the structural equations of each chain are established with the constraint relationship of the rod length as the boundary condition. The polar coordinate search method is used to conduct layer-by-layer iterative searches to determine the reachable boundary. Finally, the shape of the workspace is plotted by processing different cross-sections, as shown in Fig. 5, which illustrates the workspace for the restricted area and attitude capability of the simulator. The overall design does not interfere with motion, and the shape of the workspace undergoes a gradual change without singularities. Table 1 summarizes the operational limits of the 6-DOF simulator. The translational range in the Z direction spans from 621 mm to 985 mm, while the ranges in the X and Y directions are ±366 mm and ±422 mm, respectively. The rotational capabilities about the RX, RY and RZ directions are constrained to ±27°, ±23° and ±30°, respectively.

The simulator works under heavy load conditions, and thus, the dynamic characteristics of the six limb actuators have a considerable impact. The mass characteristics of each local component, in combination with the independent system variables that describe the virtual velocity, are instrumental in deriving the principle of virtual work. The virtual work of the inertial force equals the work performed by the external force of the system, and its dynamics equation can be deduced as follows:

$$\mathbf{M}(\mathbf{q}) \cdot \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{G}(\mathbf{q}) = \mathbf{F} \quad (3)$$

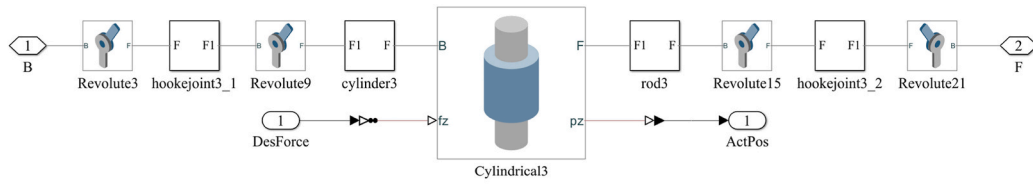
where $\mathbf{M}$ is the total mass, $\mathbf{q}$ is the position and attitude vector of the motion platform, $\mathbf{C}$ is the total inertial force, $\mathbf{G}$ is the total gravity, and $\mathbf{F}$ is driving force of the limbs.

2.3. Establishment of physical system model

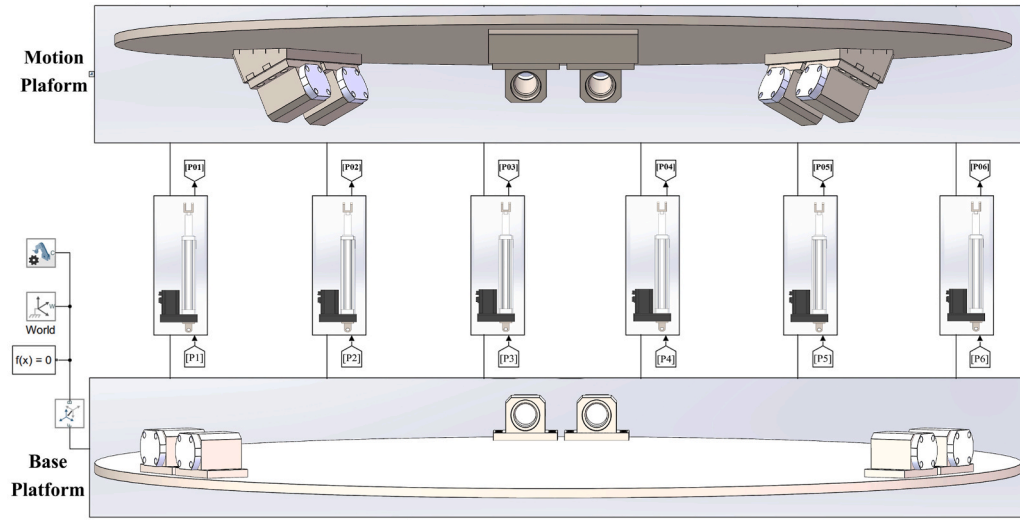
The physical system model serves as the foundation to support the subsequent multi-case input and control study, which enables determining the motion state under a specific external moment (Xia et al., 2020). The overall physical system model is constructed integrally using an MBD model, an input and inverse kinematic system model, a sensor system model, and a control system model using the Simscape tool in Matlab/Simulink, as shown in Fig. 6. The MBD model executes specific actions derived from the kinematic and dynamic analyses. The input and inverse kinematic system model updates the expected CFD-driven inputs and solves the inverse kinematics. The sensor system model tracks the pose of the components, and the control system model corrects the error and generates actual control outputs.

Fig. 6(a) and (b) show the MBD models for the whole system and a limb actuator subsystem. In each limb actuator subsystem, the motion between the limb and the hinge is set as a revolute joint, while that between the cylinder and the rod is set as a cylindrical joint through the Simscape-Multibody module. Moreover, the initial position, bound, input force and automatically computed motion of the rods are added. The limits of the spring stiffness, damping coefficient, and transition region width are set to describe the internal mechanics of the motor to restrict its velocity and acceleration. For the overall MBD model, according to the mutual kinematic relationship between components, physical properties such as the mass and inertia of the rigid body and modular properties such as joints, constraints, external forces and kinematic pairs are defined. Subsequently, the platforms and limb subsystems are connected to the rigid body module, coordinate transformation module, and kinematic pair module. Finally, an MBD model is constructed by configuring the simulation parameters and defining the initial conditions and external inputs.

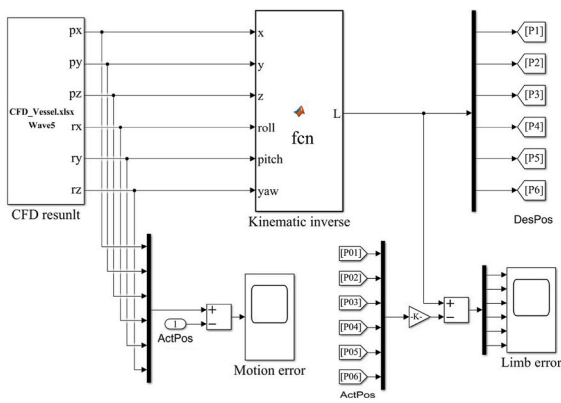
Fig. 6(c) shows the input and inverse kinematic system model. The input system model adopts the motion values obtained from CFD simulations as inputs with a certain scale reduction according to the designated workspace. The inverse kinematic system model utilizes an inverse kinematics to calculate the theoretical telescopic lengths of the six rods based on the desired pose of the motion platform.



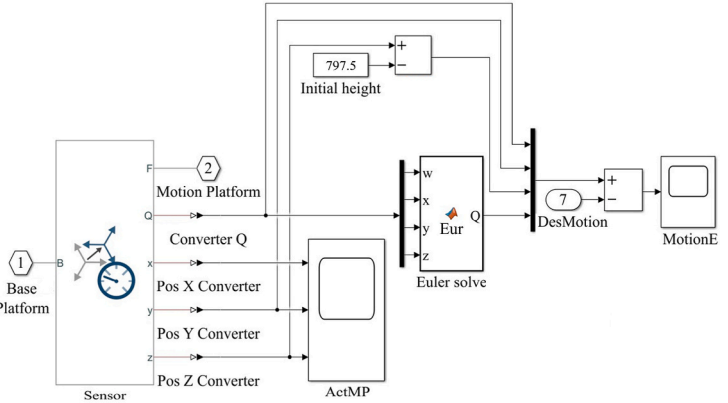
(a) The MBD model of a limb actuator subsystem



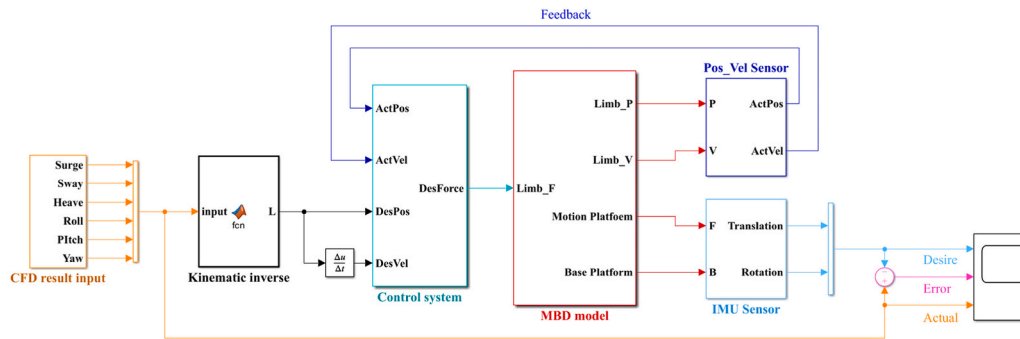
(b) The whole MBD model



(c) Input and inverse kinematic system model



(d) Sensor system model



(e) Overall physical system model

Fig. 6. Physical system model in Matlab/Simulink.

The sensor system model monitors the motion of platforms and six limb actuators, as shown in Fig. 6(d). The translation and Euler angle of the two coordinate systems are measured to assess the control effect. Additionally, the actual displacement and speed of the limbs are monitored to provide feedback data for the control system model. The control system model incorporates the errors and the change rate of the errors in the displacements of six rods as the inputs, and performs active control of each limb to output the required driving force based on the proposed advanced control strategy.

The overall physical system model is shown in Fig. 6(e). Combined with the CFD results of the extreme behavior as inputs, the pose parameters are resolved. The control signals are sent to the six limb actuators to adjust the driving forces in real-time by the controller. The motion errors of the six rods and motion platform obtained from the sensors are fed back to form a closed-loop control system. Finally, the physical system model can integrate multiple models for simulation and control, thereby optimizing the performance of the system based on the diversity of inputs and the effectiveness of control strategy.

3. CFD simulation of extreme vessel motions under multiple sea states

To construct the input system model for the physical system model, it is important to study the 6-DOF motions of a vessel in waves and incorporate the behavioral authenticity effects. The extreme behavior under complex sea states must be analyzed in the absence of the propulsion and rudder systems to obtain the diverse inputs and prior conditions for subsequent physical system model simulation and testing. This section establishes the numerical wave tanks to simulate the dynamic coupling between a vessel and waves based on the XFlow software. A multiphase CFD model that performs the numerical simulation for the vessel motion under limiting waves of multiple sea states with LBM is also constructed.

3.1. Numerical methods

Current research on vessel motion analysis focus on classical finite volume and finite element methods grounded in the Navier-Stokes equations. The LBM tackles critical limitations in traditional CFD methods (Watanabe and Aoki, 2021; Watanabe et al., 2023): the meshing process is removed as the simulation relies on an automatically generated lattice, LBM scheme uses a large eddy simulation (LES) turbulence model at low computational cost, and mathematical formulation of the Boltzmann transport equation is straightforward to resolve. Consequently, the numerical simulations are performed using the XFlow software with the LBM, LES model, and wall treatment method.

1) Lattice-Boltzmann method

LBM schemes are based on minimal kinetic models for the Boltzmann equation, used with adaptive lattice and flow field reconstruction techniques to capture turbulence-scale information efficiently (Chu et al., 2016). The LBM model is solved numerically using a time-stepping model based on a propagate-collide scheme with a lattice discretization. In the continuum space with discrete velocities, the Boltzmann transport equation can be discretized on the lattice as:

$$\frac{\partial f_i}{\partial t} + \mathbf{e}_i \cdot \nabla f_i = \Omega_i \quad (4)$$

$$f_i(\mathbf{r} + \mathbf{e}_i, t + dt) = f_i(\mathbf{r}, t) + \Omega(f_i) \quad (5)$$

where f_i is the probability distribution function, $\mathbf{e}_i$ is a discrete set of velocities, and $\Omega(f_i)$ is the collision operator that computes a post-collision state-conserving mass and linear momentum.

As a discrete approximation of the continuous Boltzmann equation,

Table 2

Main parameters of a full-scale vessel.

Length (m)	Width (m)	Height (m)	Draft (m)	Displacement (t)
57.75	10.80	6.25	3.60	1000

the streaming or propagation step models the advection of the particle distribution functions along discrete directions, and the collision operator models most of the physical phenomena. Macroscopic variables such as the density, velocity and pressure are obtained by sampling ample particles and derived from the statistical moment:

$$\rho = \sum_{i=1}^b f_i \quad (6)$$

$$\rho \mathbf{v} = \sum_{i=1}^b f_i \mathbf{e}_i \quad (7)$$

Here, ρ is the density of the fluid and $\mathbf{v}$ is the particle velocity vector.

2) Turbulence model and wall treatment method

The LES turbulence model proposes a satisfactory intermediate solution between the Reynolds-averaged Navier-Stokes and direct numerical simulations. Most of the turbulence is directly resolved, while smaller scales are modeled to reach accessible computation times. This approach is beneficial for wave transient simulations that require high turbulence and separated flow to more accurately predict turbulent structures, turbulence-wall interactions, and the eddies.

The wall adapting local eddy (WALE) viscosity model adjusts the sub-grid local turbulence parameters to provide a consistent local eddy viscosity and near-wall behavior. It introduces an additional turbulent eddy viscosity ν_t to offer advanced turbulence prediction in the near-wall area and boundary layer. The actual execution is formulated as follows:

$$\nu_t = \Delta_f^2 \frac{(G_{\alpha\beta}^d G_{\alpha\beta}^d)^{3/2}}{(S_{\alpha\beta} S_{\alpha\beta})^{5/2} + (G_{\alpha\beta}^d G_{\alpha\beta}^d)^{5/4}} \quad (8)$$

where Δ_f^2 is the filter scale and $S_{\alpha\beta}$ is the strain rate tensor of the resolved scales.

3.2. Setup details

3.2.1. Computational domain and boundary conditions

A numerical wave tank gas-liquid-solid multiphase coupled model of a full-scale vessel is constructed. The structural parameters of the vessel are shown in Table 2. The vessel is configured to adhere to rigid-body dynamics, moving forward at a low velocity while maintaining freedom in all six directions. When combined with mechanical (full-load mass and moment of inertia) and interaction properties (restitution coefficient, static friction coefficient, and dynamic friction coefficient), the simulations should maintain the vessel at a certain draft depth and monitor the coordinate of the mass center.

Given the vessel dimensions and real navigational environment, the fluid computational domain and boundary conditions are established in Fig. 7. Combined with the verification of the boundary scale independence, the length, width and height of the computational domain are designated as $6L$, L , and $2L$, respectively, where L is the length of the vessel. Specifically, the fluid domain is established at the front end of the vessel as $2.5L$, and the sea depth on the free surface is $1.2L$. For boundary conditions, the left and right surfaces of the fluid domain are defined as the velocity inlet and outlet, respectively, such that the fluid flows through the hull in a reverse direction. The bottom and side surfaces are set as walls with enhanced wall function, while the top surface is

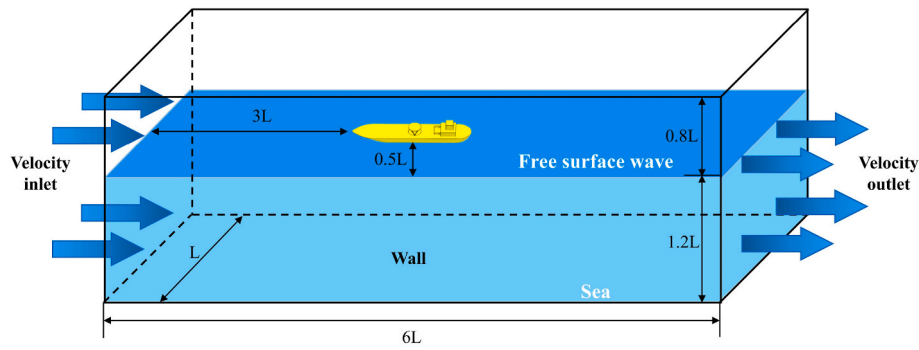


Fig. 7. Computational domain and boundary conditions.

Table 3

Limit-wave parameters of free surface for multiple sea states.

Sea state	Wind speed(m/s)	Wave height(m)	Cycle(s)
Level 3	5.5	1.25	3.8
Level 4	7.9	2.5	5.4
Level 5	10.8	4	7.0

configured as a pressure outlet.

3.2.2. Numerical wave tank

CFD is employed to construct the limit wave motions based on extreme values corresponding to wave parameters under each sea state. Table 3 presents the adopted limit-wave parameters for various sea states from the International Sea and Swell Scale. To attain a higher-order approximation of the morphology and characteristics of target waves, the amplitude and frequency of progressive waves are determined based on the fifth-order Stokes expansion of small-amplitude waves. The wave velocity and consistent laws are automatically accounted for and calculated in the XFlow wave setup methodology to describe the initial free surface and the inlet wave function. At the same time, the gravitational potential, free surface tension, and viscous interaction are enabled in the free surface flow model to set up the

simulation of fifth-order Stokes surface gravity waves at the light (air) and heavy fluid (sea water) interfaces. Fig. 8 shows the initial velocity and pressure distributions in the numerical free-surface case. The velocity of the free surface varies with wave height, and the pressure changes in a gradient with sea depth, which simplify the free-surface flow setup and retain the physical fidelity for waves. Since the hydrostatic pressure is significantly higher than the initial wave-dynamic pressure, a linear pressure distribution assumption based on the hydrostatic pressure formula aligns with the initial conditions of the Stokes wave.

The resolved scale of the global lattice structure is set to 0.5 m. To calculate the effect of fluid-structure coupling, it is necessary to track the free surface with sufficient resolution. The local lattice structure for an

Table 4

CFD solution parameters.

Solution parameters and tools	Solving setup
Flow model	Free surface
Turbulence	WALE
Initial conditions of global attributes	Water channel default
Force evaluation scheme	Surface stress integration
Refinement algorithm	Adaptive refinement

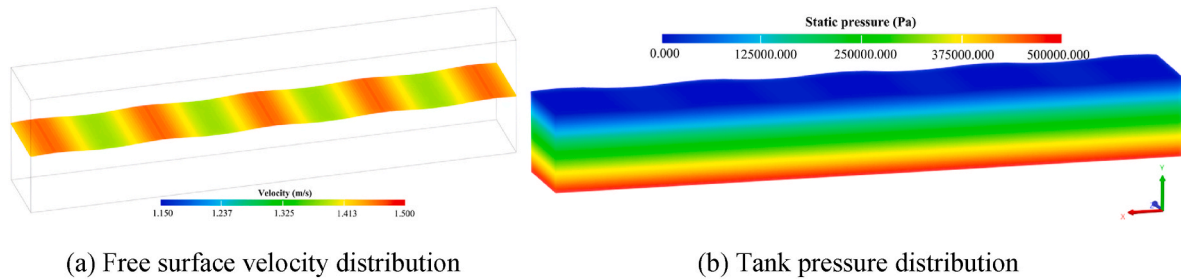


Fig. 8. Numerical wave tank.

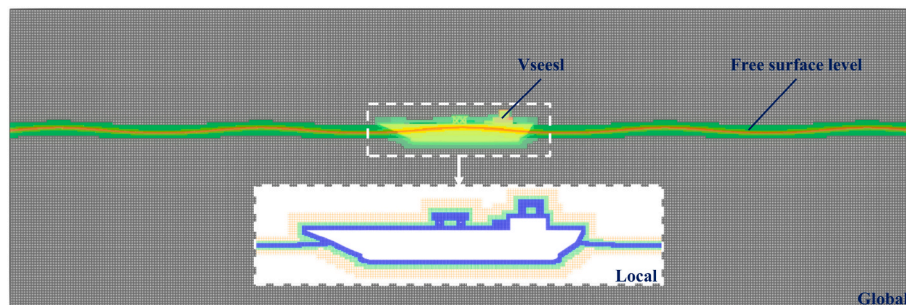


Fig. 9. Domain lattice structure.

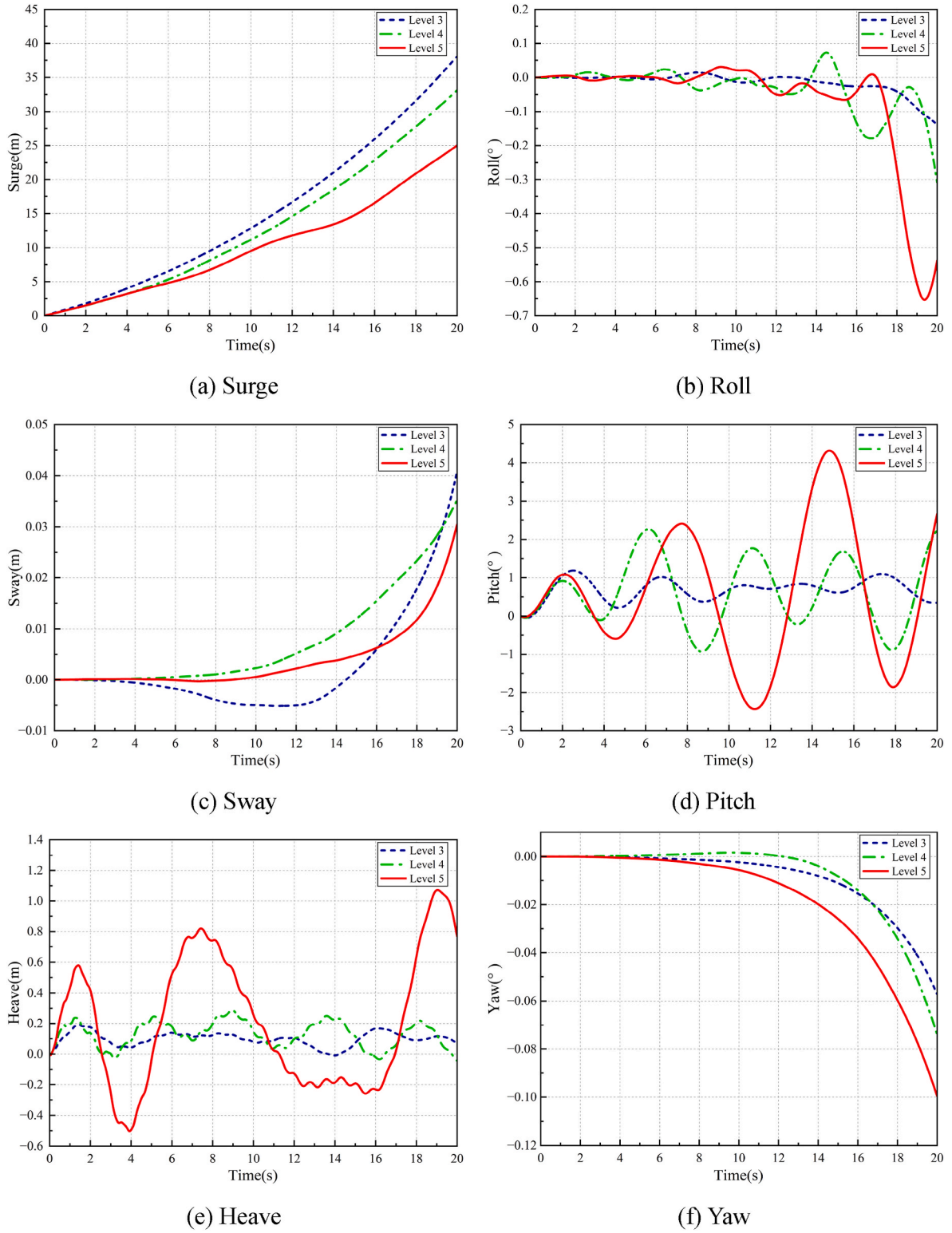


Fig. 10. Extreme vessel motions under various sea states.

initial 0.125 m interface resolution is adaptively refined around the free surface wake and moving vessel hull according to the flow characteristics. Fig. 9 provides the global and local domain lattice structures. The lattice structures are locally adjusted to follow the wave propagation and diffusion in different regions and better capture the intricate interaction changes.

The CFD solution parameters and tools are listed in Table 4. The free

surface is constructed by setting progressive wave boundary conditions based on the initial conditions of the default water channel. Turbulence closure is achieved via the WALE viscosity model within the LES. The force evaluation scheme with surface stress integration is adopted to calculate the total force on the vessel. Volume correction is enabled to ensure fluid volume convergence. Finally, considering the repetitive and cumulative effects of wave forces on the vessel under multiple wave

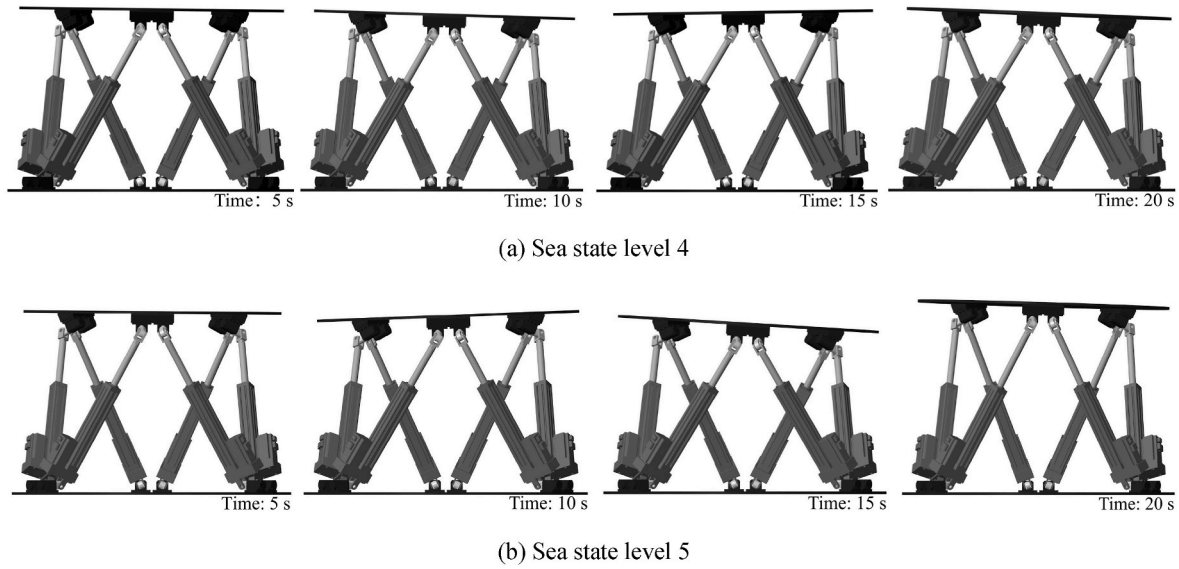


Fig. 11. Physical system model simulation under sea state levels 4 and 5.

cycles, transient simulations are performed for 20 s with a 0.0018 s time step based on balancing the computational accuracy and efficiency.

3.3. CFD numerical results

A sailing yacht hull was used to assess the influence of the parameters on the hydrodynamic performance (resistance, seakeeping etc.), which validated the XFlow free surface engine results (Stoker, 1957; Wu et al., 2018). In this study, the pressure and velocity fields around the vessel are obtained by solving the fluid force and fluid moment of the rigid body system. Then the motion equations of the vessel with the wave motion are calculated, as to obtain the curves of the 6-DOF extreme motion of a vessel under multiple sea states, as shown in Fig. 10.

Fig. 10 illustrates the 6-DOF vessel motions over time under various sea states. In sea states ranging from 3 to 5, after a transient period, the heave motion exhibits periodic fluctuations at 0.08 m above the mass center. As the sea state level increases, the increased wave height leads to increasing resistance of the vessel. Moreover, the interaction and turbulence effect also become more significant, thereby making the 6-DOF vessel movement more intense.

The results reveal that the vessel experiences significant heave, sway, and surge motions when navigating against waves. The amplitudes of heave motion under sea state levels from 3 to 5, measuring 0.1124, 0.2077, and 0.9914 m, respectively, far exceeded those of sway motion. The maximum values of roll motion under different sea states are 0.148°, 0.307°, and 0.662°, while the pitch amplitudes are 1.18°, 2.27°, and 4.32°, all of which exceed the amplitudes of yaw motion, in line with the actual movement. In fifth-level sea state, heave motion resulted from the disparity of wave height, and pressure leads to a change in vessel draught. Pitch motion arose from variations in fore and aft waves influences the undulation of the hull. Both collectively impact the navigational performance. Therefore, the necessity to assess the impact of extreme vessel motions on shipboard equipment becomes paramount across various sea states.

In summary, the 6-DOF extreme behavior of a vessel under multiple sea states predicted by the CFD methods serves as critical inputs for the physical system model simulation and test. Moreover, modifying vessel parameters and sea state levels yields a diverse CFD-driven input system model, providing complex simulation situation in a laboratory environment.

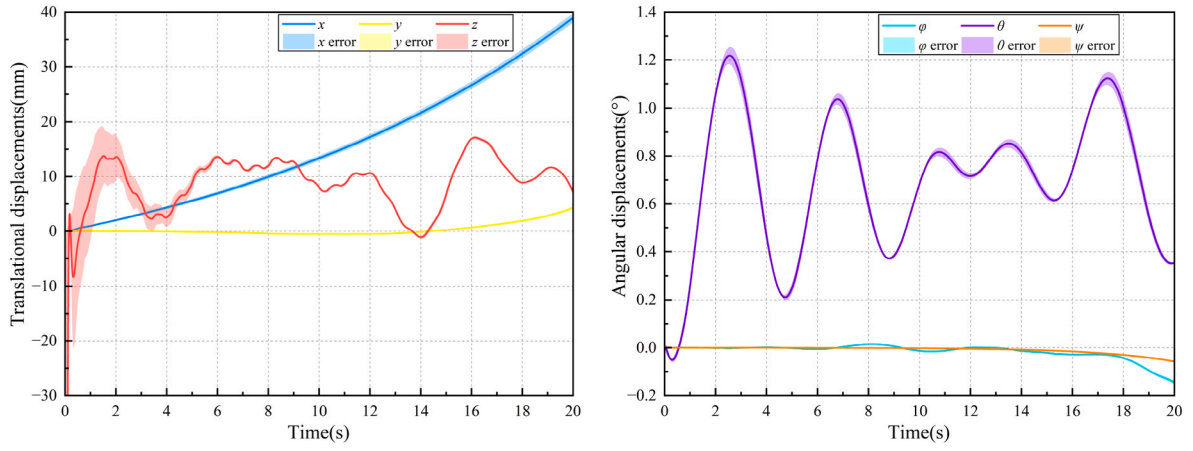
4. Analysis of physical system model simulation and test

Leveraging the behavior of a vessel in waves through the CFD simulation, the simulator adopts a mechanical approach to reproduce various motions and simplifies the real experimental condition in both the actual system and the physical system model. This section combines physical system model simulations and tests with inputs from CFD results to illustrate the system performance and the accuracy of the physical system model. It will provide a prerequisite for the subsequent analysis of the performance of control strategies.

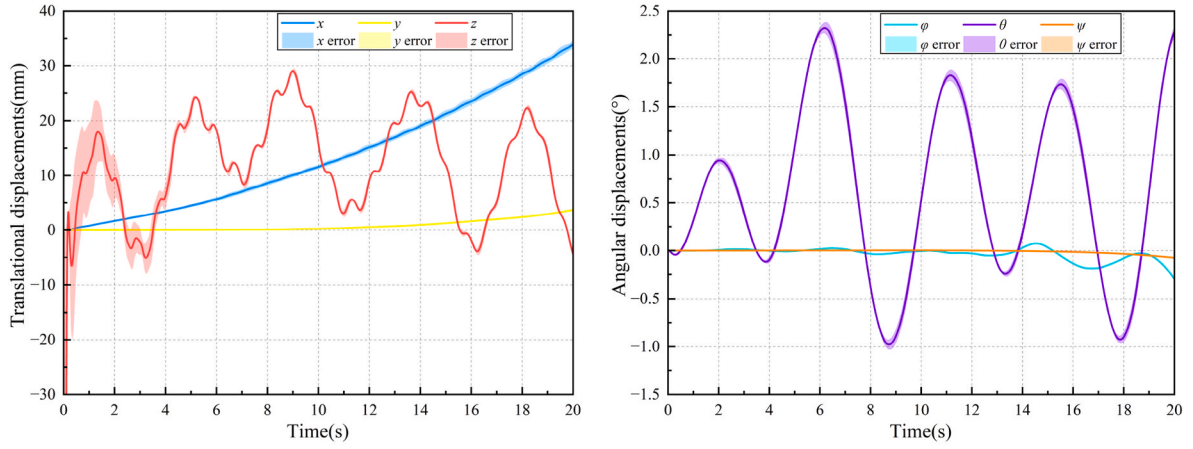
4.1. Physical system model simulation

To demonstrate the performance of the physical system model, motion simulations for the 6-DOF simulator (surge, sway, heave, roll, pitch and yaw) are conducted. According to the solved workspace, the input system model takes the CFD motion values of a vessel in multiple sea states as inputs in an appropriate ratio. For shipboard equipment such as the wave compensation system, a load of 12,000 N is applied to the center of the motion platform. To actualize the Simulink simulation of the physical system model, the pose of the motion platform is accomplished by coordinated control of the six limbs through the inverse kinematics solution. The initial state is shown in Fig. 3(b), and the simulation at different moments under sea state levels 4 and 5 are shown in Fig. 11.

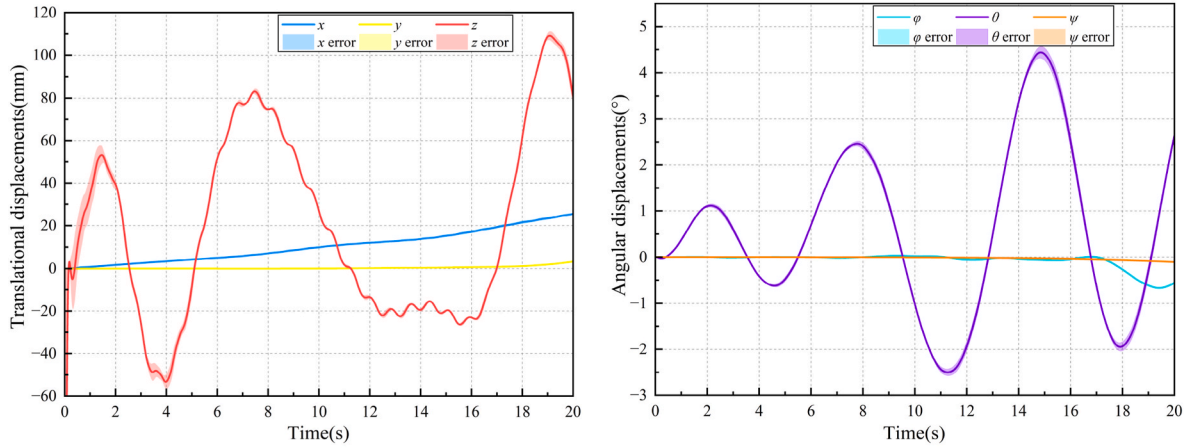
The results in Fig. 12 show the translational and angular displacements of the motion platform and their respective error bands with time for sea state levels 3 to 5. It is evident that the displacement results are consistent with theoretical inputs, substantiating the reliability of the kinematic and dynamic analyses. As the sea state level increases, the motion of the vessel becomes more violent, and the telescopic variation range of the rods also increases, accompanied by an increase in errors. The motion platform moves to the neutral height position at the beginning. Due to the abrupt application of the driving force and the effect of the heavy load, the system engenders an unstable dynamic state. This leads to a large absolute value of the translational displacement error in the y direction, and rapidly reduces and tends to fluctuate within a stable range. Moreover, under relatively stable conditions, when there are significant changes in the telescopic direction and instantaneous displacement, the limitation on the rod acceleration leads to more errors in the translational displacement along the z-axis and the rotational displacement around the y-axis.



(a) Sea state level 3



(b) Sea state level 4



(c) Sea state level 5

Fig. 12. Motion curve and error of the motion platform.

Since the goal is to maintain the motion platform in accordance with the desired inputs, the global coordinates of the position and attitude deviation are used as motion errors to evaluate the performance of the simulator. The maximum deviations of the motion platform for different sea states are shown in Table 5. When the motion is stable, the maximum values of 6-DOF deviation are 0.87 mm, 0.24 mm, 5.10 mm, 0.02°, 0.16°

and 0.001°, which are relatively small compared to the overall motion scale. Furthermore, the maximum errors attribute to the telescopic displacement of the rods and local motion fluctuations due to the mutual coupling effect caused by heavy loading, which need to be controlled further in the subsequent steps. The error in vessel motion obtained by the overall physical system model is relatively small, which verifies the

Table 5
Simulation error maximum of the motion platform.

	Sea state level 3	Sea state level 4	Sea state level 5
x (mm)	0.8698	0.8091	0.6978
y (mm)	0.2161	0.1670	0.2363
z (mm)	2.0365	2.9168	5.0972
φ (°)	0.0072	0.0111	0.0196
θ (°)	0.0044	0.0901	0.1645
ψ (°)	0.0012	0.0012	0.0012

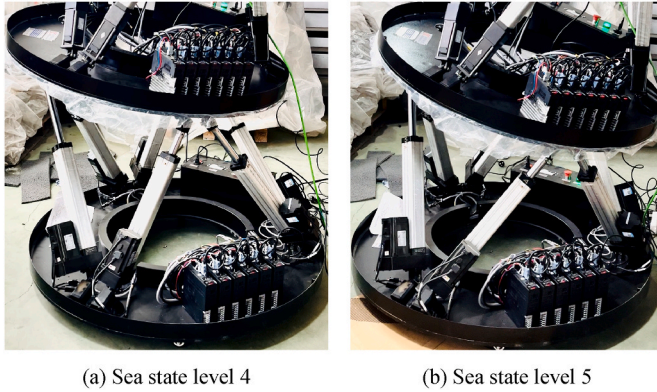


Fig. 13. Actual test of the simulator.

feasibility of the MBD model.

The physical system model is constructed integrally to enable component behavior-based simulation of system performance. The simulation results demonstrate that the physical system model exhibits excellent motion performance, tracking preset targets accurately and implementing pose and state feedback effectively. The proposed modeling and simulation method for a vessel motion simulator is considered feasible and can provide valuable support for subsequent research.

4.2. Test verification

The experimental system of the vessel motion simulator conducts comprehensive performance tests, relying on data input, kinematics inverse solution, motion control, real-time measurement, and evaluation. Fig. 13 shows the actual pose of the simulator in the operational test under sea state levels 4 and 5. The initial system is at median height, and the same load is applied at the center of the motion platform. After the obtained behavior from CFD simulations is solved for the inverse kinematic solution, it is sent to the system through user datagram protocol (UDP) communication. When the system implements the execution of actions in the test operation, an inertial measurement unit (model: LPMS-BE-SM, angular resolution $<0.001^\circ$) is generally used to measure and resolve the 6-DOF actual pose of the motion platform. The errors for the motion platform between the desired inputs and the actual pose is calculated to evaluate its performance. Integrated with the telescopic displacement monitored by displacement sensors, further feedback is provided to the control system to form a closed-loop control to adjust the

driving force and torque in real-time. The control system corrects the telescoping deviation of six rods, and the motion platform can remain in motion in six degrees of freedom.

The average errors of the experimental motion under different sea state tests are obtained, and a comparative analysis with the simulation results is shown in Table 6. The error is marginally larger than that of the Simulink result, but the deviation between the two is relatively small. The deviation arises from two aspects, one is the hysteresis caused by the communication delay, and the other is the mechanical error, including the processing and installation errors of each component. Due to the 1 ms delay inherent to UDP communication and the friction in the mechanical structure, the motion of the platform exhibits a lag behind the desired motion. Moreover, the heave range and its hysteresis are more obviously pronounced in comparison to the other degrees of freedom. For some local changes in the motion with larger amplitudes, the constraints on the motor speed and acceleration lead to the requirement of a sufficient response time.

The vessel motion responses to various sea waves are slightly random due to the randomness of the actual waves. The simulator is only required to generate motions in various degrees of freedom and their compound motions under specific amplitudes and periods. From the comparison in Table 6, it is seen that the physical system model and actual system of the 6-DOF simulator can meet the requirements for simulating vessel motions in different sea conditions.

Fig. 14 presents the test displacements and errors of six rods from sea state levels 3 to 5. It can be seen that the simulator is capable of simulating motion in accordance with a predetermined trajectory and the trends in the displacement errors of the six rods are consistent. In sea states levels 3 and 4, the displacement errors are located within the range of 7 and 14 mm. Some values slightly higher in the steady state due to the coupling effects in the mechanism and communication hysteresis. In level 5, the range of motion increases substantially and the stroke of the rods changes significantly; the corresponding maximum error reaches up to 58 mm. From the perspective of limbs, their telescopic error becomes particularly pronounced in heavy loads and large-scale motions. Therefore, it is important to introduce a superior control strategy to compensate for the errors to enhance its precision and reliability.

5. Exploration of control strategy

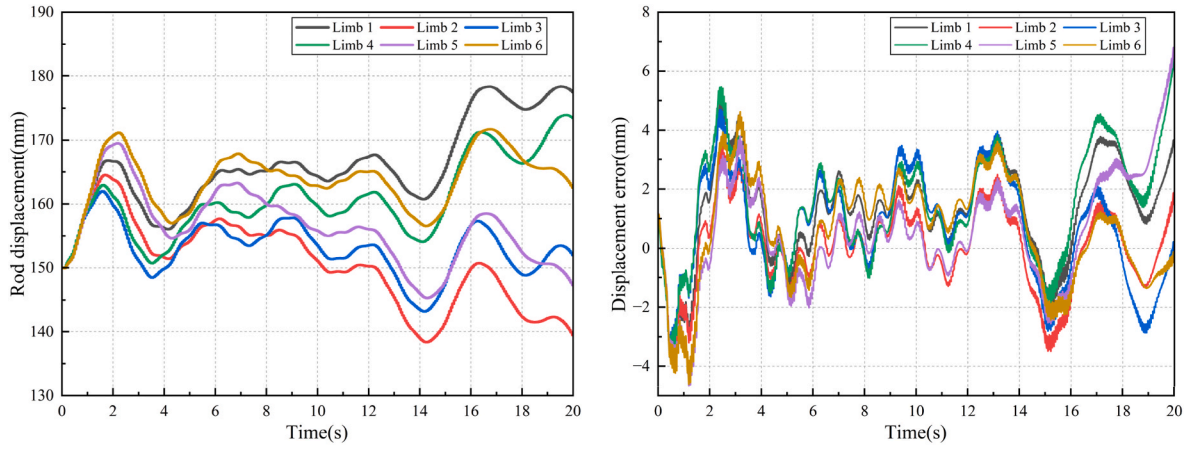
Considering the motion error in the simulation and testing of the simulator, the control system model with a well-designed control strategy is the key factor to attain the expected performance. Fuzzy PID control is widely used in marine robotic fields, but the design of fuzzy rules is generally influenced by the designers' limited experience. The introduction of optimization algorithms for fuzzy rules can significantly eliminate the influence of subjective factors.

5.1. Design of fuzzy PID controller

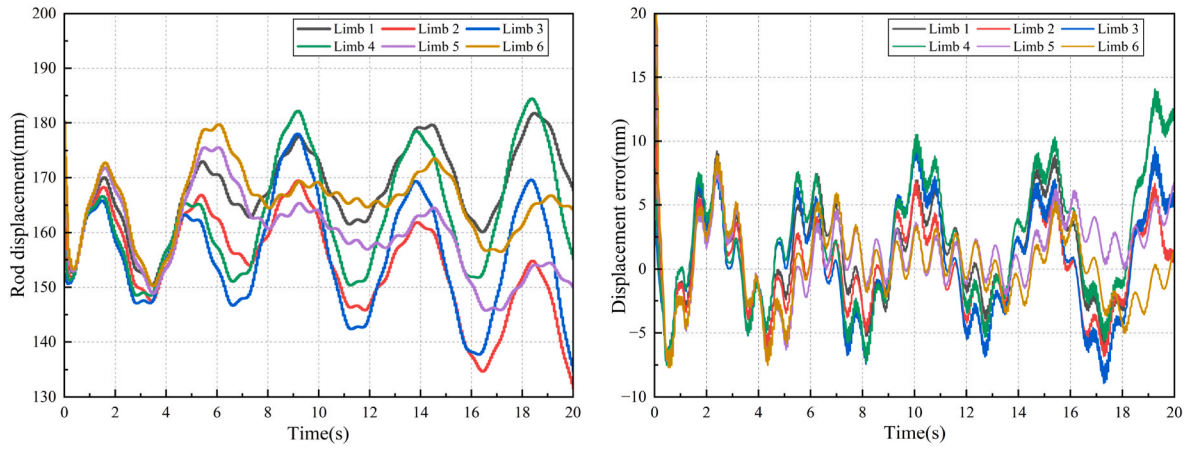
The Stewart parallel mechanism is a multi-input, multi-output and coupled system. The six limb actuators require independent control, but it is difficult to construct an accurate mathematical model because of the strong nonlinearity of intricate structures and mechanical friction. An

Table 6
Experimental and simulation errors for the motion platform.

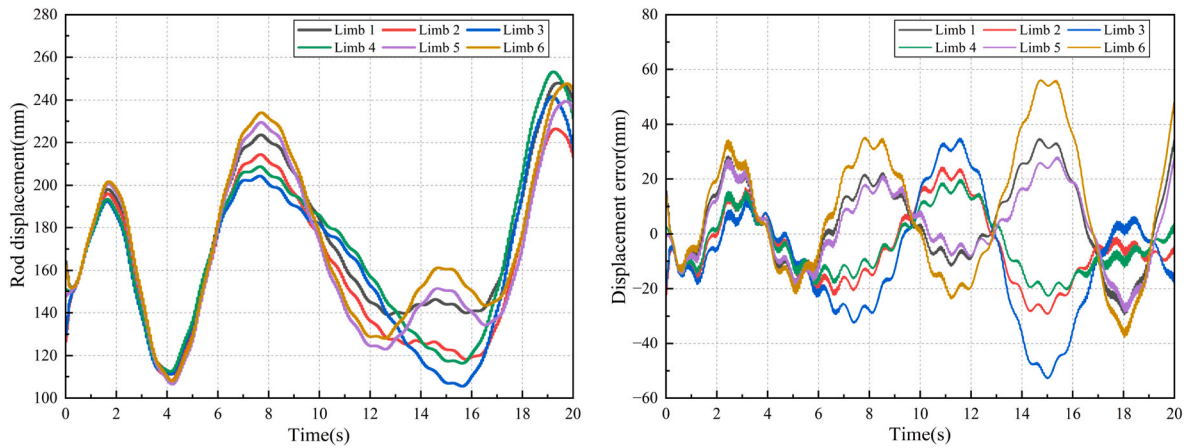
Sea state		X (mm)	Y (mm)	Z (mm)	RX (°)	RY (°)	RZ (°)
Level 3	Simulation	0.4859	0.0392	1.1058	0.0011	0.0166	0.0001
	Test	0.5359	0.0686	1.3587	0.0030	0.0809	0.0008
Level 4	Simulation	0.4298	0.0331	2.4939	0.0026	0.0428	0.0001
	Test	0.5085	0.0578	3.2968	0.0140	0.2969	0.0012
Level 5	Simulation	0.3540	0.0288	2.7416	0.0029	0.0637	0.0002
	Test	0.4057	0.0358	5.6125	0.0148	0.4792	0.0016



(a) Rod displacement and error under sea state level 3



(b) Rod displacement and error under sea state level 4



(c) Rod displacement and error under sea state level 5

Fig. 14. Experimental displacement and error of six rods.

active control system model takes the displacement of six limbs as the reference and solves for the output driving force and moment to stably meet the desired value by correcting the errors.

It is difficult to maintain optimal control efficacy in traditional PID controllers for the Stewart mechanism, and its linear combination frequently leads to a conflict between the response rapidity and

overshoot magnitude. The nonlinear approximation of fuzzy control eliminates the need of an exact mathematical model and parameter time-varying defects. Hence, the gain-tuned fuzzy PID controller is emerged, which applies fuzzy mapping to the PID gain coefficients to form a real-time adaptive adjustment.

A Mamdani-style fuzzy PID controller is designed for the motion

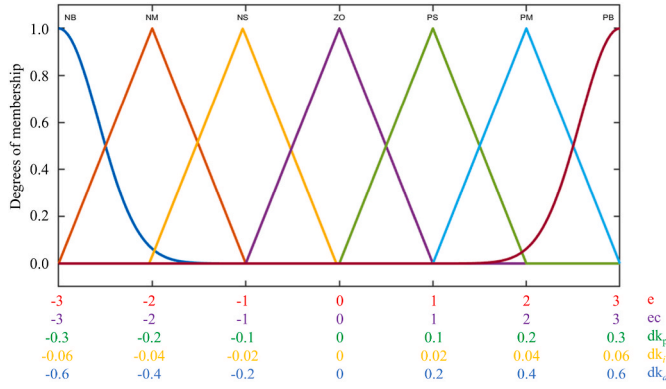


Fig. 15. Membership functions.

Table 7

Fuzzy rules of $dk_p/dk_i/dk_d$.

$dk_p/dk_i/dk_d$		ec						
		NB	NM	NS	ZO	PS	PM	PB
e	NB	PB/	PB/	PM/	PM/	PS/	ZO/	ZO/
		NB/	NB/	NM/	NM/	NS/NS	ZO/	ZO/
		NB	NB	NM	NM	NS	NS	ZO
	NM	PB/	PB/	PM/	PS/	PS/	ZO/	NS/
		NB/	NB/	NM/	NS/	NS/NS	ZO/	ZO/
		NB	NB	NM	NM	NS	NS	ZO
	NS	PM/	PM/	PM/	PS/	ZO/	NS/	NS/
		NB/	NM/	NS/NS	NS/NS	ZO/	PS/PS	PS/PS
		NB	NM	NS/NS	NS/NS	ZO	PS/PS	PS/PS
	ZO	PM/	PM/	PS/	ZO/	NS/	NM/	NM/
		NM/	NM/	NS/NS	ZO/	PS/PS	PM/	PM/
		NM	NM	NS	ZO	PM	PM	PM
	PS	PS/	PS/	ZO/	NS/	NS/	NM/	NM/
		NM/	NS/NS	ZO/	PS/PS	PS/PS	PM/	PB/
		NS	ZO	ZO	PS	PM	PM	PM
	PM	PS/	ZO/	NS/	NM/	NM/	NM/	NB/
		ZO/	ZO/	PS/PS	PS/PS	PM/	PB/PS	PB/
		ZO	ZO	PS	PM	PM	PM	PM
	PB	ZO/	ZO/	NM/	NM/	NB/	NB/	NB/
		ZO/	ZO/	PS/PS	PM/	PM/	PB/PB	PB/PB
		ZO	ZO	PM	PM	PM	PM	PM

control, where the input parameters are the rod displacement error e and its change rate ec , and the output parameters are the three parameter factors dk_p , dk_i , dk_d of the regulated PID control. The fundamental domains of e , ec , dk_p , dk_i and dk_d are $[-3,3]$, $[-3,3]$, $[-0.3,0.3]$, $[-0.06,0.06]$ and $[-0.6,0.6]$, and all have seven language variables defined as NB, NM, NS, Z, PS, PM, and PB. Fig. 15 illustrates the degrees of the membership function with the Trimf function for the middle part and the Guess function applied on both sides.

When both e and ec are small, slight positive increments in dk_p and dk_i are set to improve the stability and k_d is more massive appropriately for anti-interference capacity and to prevent oscillation. When e and ec are medium, a negative increment for dk_p is added to k_p to reduce overshoot. Similarly, when e and ec are large, k_p is supposed to increase and k_d decreases, while k_i tends to be zero to avoid significant overshoot (Dai et al., 2020). Based on empirical experience, the fuzzy rules are established in Table 7, and the centroid strategy is applied in defuzzification.

5.2. Optimization of fuzzy PID controller

Inappropriate design of fuzzy rules may result in the degradation of control performance, and the robustness remains may not be optimum. It is valuable to introduce a GA to optimize fuzzy rules for higher control accuracy and stability. The displacement errors are used as an index to calculate the individual fitness through the simulation. GA completes

the population evolution and obtains the best adaptation degree of the online parameters. The optimization flow chart is shown in Fig. 16.

In this study, a total of 147 parameters are optimized, as listed in Table 7. All parameters are integers and in the range $[1,7]$, which is equivalent to seven language variables. GA always minimizes the displacement error in the movement, and the objective function is given by:

$$\min f(e) = \sqrt{\frac{\sum_{i=1}^n e_i^2}{n}} \quad (9)$$

The selection operation uses a roulette wheel with a playback-based random sampling method to select superior individuals from the existing population as breeding parents. The crossover and mutation operators are the most critical parameters of the GA, which collaborate for the global exploration and local exploitation.

1) Crossover operator

The crossover operator generates several new individuals and prevents the destruction of valuable individuals. The crossover probability takes the value 0.9 to boost the convergence speed. This study adopts a parent-centered Laplace crossover (LX) operator to realize strong crossover in the gradual stage and weak crossover in the convergence stage. The random number β that follows a Laplace distribution can modulate the density function coefficient to regulate the offspring population as

$$\beta = a + \text{sign}(u - 0.5) \cdot b \cdot \ln(1 - 2|u - 0.5|) \quad (10)$$

where u are random numbers distributed within the interval $[0,1]$ and a and b are the Laplace distribution parameters. As the parameter b approaches 0.5, the new individuals better approximate the parent individuals.

The new individual obtained after the crossover operation between the two parents is given by:

$$X_A^{k+1} = \beta X_A^k + (1 - \beta) X_B^k \quad (11)$$

$$X_B^{k+1} = (1 - \beta) X_A^k + \beta X_B^k \quad (12)$$

2) Mutation operator

The mutation operator contributes to the population diversity, generates new genotypes, and prevents getting trapped in local optima. A mutation operator based on power mutation (PM) is employed to introduce a larger mutation range. According to a 20-group standard global optimization test, the PM operator in conjunction with the LX operator possesses an excellent global optimization ability (Deep and Thakur, 2007). The offspring individual is generated by the following process:

$$X^{k+1} = \begin{cases} X^k - s(X^k - x^l), \frac{X^k - x}{X^k - x} < r \\ X^k - s(X^k - x^u), \frac{X^k - x}{X^k - x} \geq r \end{cases} \quad (13)$$

where $r \in [0,1]$ is a random number that follows the PM density distribution, and x^l and x^u are the upper and lower boundaries of the variable, respectively.

The initial population size is set to 200. During the simulation, the GA updates the control parameters according to the fitness function values in each iteration until it meets the iteration conditions and gives the best parameter value. After 200 generations, the optimal parameters are acquired, and optimized fuzzy rules are shown in Table 8.

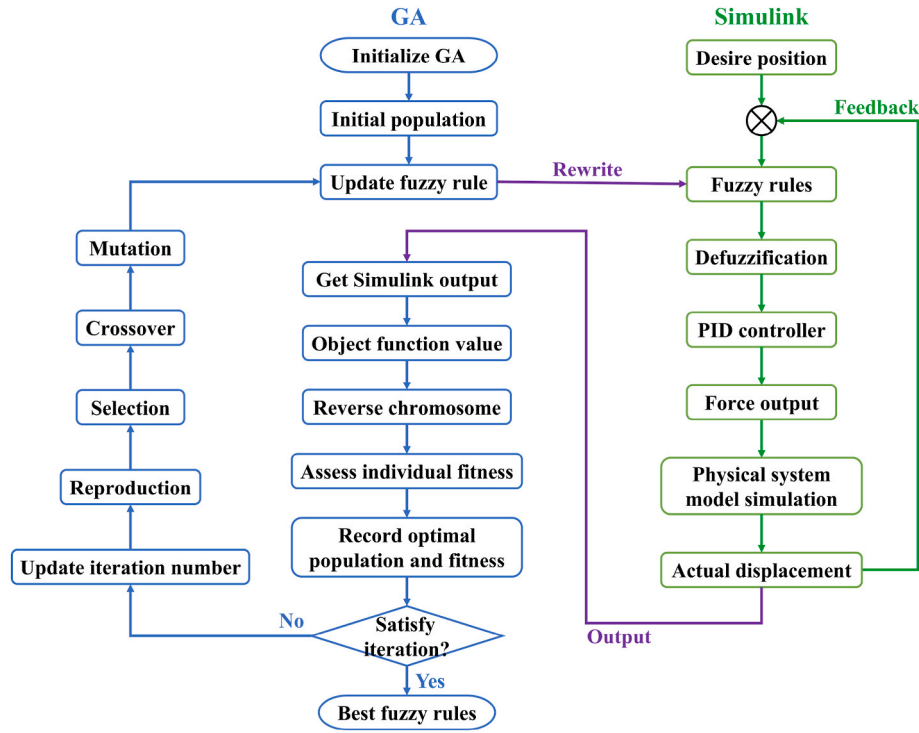


Fig. 16. Optimization flow chart.

Table 8
The optimized fuzzy rules.

	$\frac{dk_p/dk_i}{dk_d}$	ec						
			NB	NM	NS	ZO	PS	PB
e	NB	NB/	NS/	ZO/	NM/	NM/	NB/	PS/
		PS/	NM/	PM/	NB/	NM/	PB/NS	NM/
		NM	NB	PS	NB	PS		NB
	NM	NB/	NB/	PM/	PB/	ZO/	NS/	NS/
		NB/	NS/NS	PM/	NB/	NS/	PS/PB	PB/
		PB	PS	NM	PM			NM
	NS	NS/	PB/	ZO/	PS/	NB/	NM/	ZO/
		NM/	NB/NS	NS/NS	NM/	NS/	ZO/	PM/
		NS		ZO	NM	ZO		NM
	ZO	ZO/	NM/	NS/	PB/	PB/	ZO/	PS/
		NB/	NM/	NB/	PS/NS	PS/ZO	PM/	PS/
		ZO	NS	NM		NB		PM
	PS	PS/	ZO/	PB/	ZO/	ZO/	ZO/	PM/
		PS/	PS/	PS/NB	PB/	PB/NS	PB/NS	PS/PS
		NM	NM		NM			
	PM	PM/	PB/	NS/	NM/	NM/	PM/	PM/
		PM/	NM/	ZO/	NM/	PS/NB	PS/	ZO/PB
		PS	NB	NM	NB		NM	
	PB	PB/	ZO/	NM/	NM/	PS/	NM/	PS/
		PB/PB	NM/	PB/NS	PB/ZO	NB/	NB/PS	PB/
			ZO			NB		PM

5.3. Control simulation results and discussions

After the fuzzy PID control optimized by the GA is obtained, motion simulations of the physical system model are conducted to evaluate the comprehensive control performance. Considering the multi-level sea conditions, we select the PID and fuzzy PID control methods to compare the performance of the proposed GA-fuzzy PID control strategy for motion accuracy, response time, and robustness.

Fig. 17 displays the 6-DOF motion deviation of the motion platform corresponding to the desired input under sea state level 5, using the PID algorithm, fuzzy PID algorithm, and the GA-fuzzy PID algorithm, respectively. It is found that all the controllers are capable of motion

tracking, but the GA-fuzzy PID controller has better control accuracy. In the simulation process, the driving force and displacement of the MBD model are continuously adjusted by the proposed controller to correct the deviations. The response speed of the GA-fuzzy PID controller is better than those of the other controllers, and less fluctuation occurs under sudden disturbances.

The larger the proportional gain of the original fuzzy PID controller, the faster the response speed of actuators, thus improving the tracking performance and reducing the 6-DOF residual motion. However, based on the empirical fuzzy rule, higher response speeds of the actuator introduce an obvious overshoot and oscillation, resulting in poor stability. Furthermore, the residual motion under the GA-fuzzy PID controller is smoother than that under other controllers. In fact, the PID controller eliminates errors through feedback, and its control lags behind the error generation. The proposed controller can predict and suppress the speed oscillation owing to internal parameter perturbation or external load disturbance, and use feedback to compensate for the deviation to overcome the contradiction between the inherent rapidity and stability.

Table 9 shows the 6-DOF pose errors of the motion platform under the load disturbance with each control strategy. It can be seen that all algorithms have a good compensation effect for the motion disturbance. The average values of the 6-DOF residual motion with the original fuzzy PID controller are 0.4835 mm, 0.0404 mm, 2.3312 mm, 0.0038°, 0.0552°, and 0.0006°, while those with the GA-fuzzy PID controller are 0.4797 mm, 0.0378 mm, 2.2672 mm, 0.0024°, 0.0487°, and 0.002°. The errors with GA-fuzzy PID controller are smaller than those of other controllers, especially for the sway, heave, and pitch motions. The proposed control strategy has a better control performance in both position deviation and attitude deviation for diversified inputs, which is consistent with the above conclusion.

Fig. 18 presents the displacement error of limb 3 with three control strategies. The GA-fuzzy PID controller achieves the desired displacement under various internal and external disturbances and compensate for the disturbance to track the trajectory. In contrast, the PID controller obtains poor performance only through negative error feedback. The

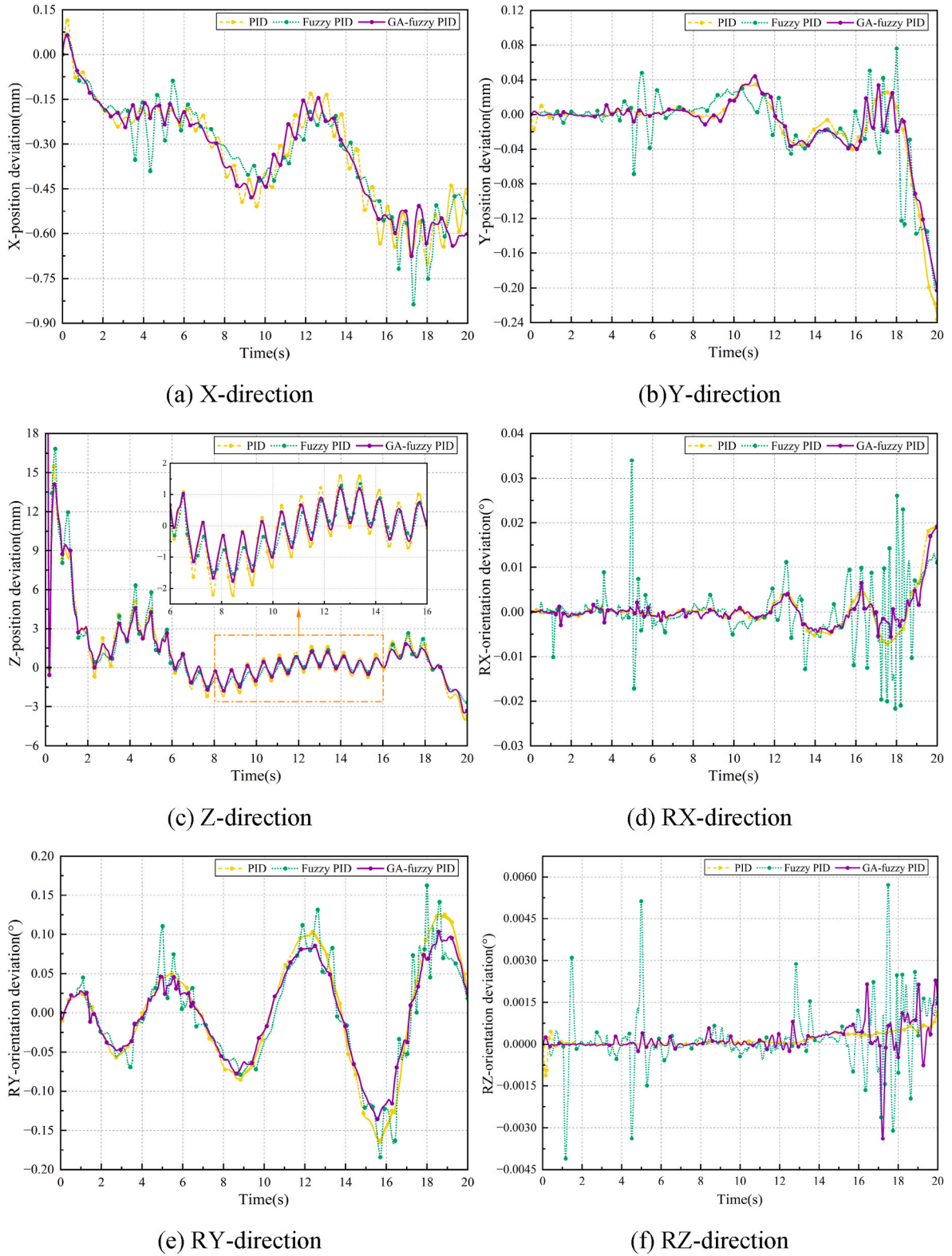


Fig. 17. 6-DOF motion deviation of the motion platform under sea state level 5.

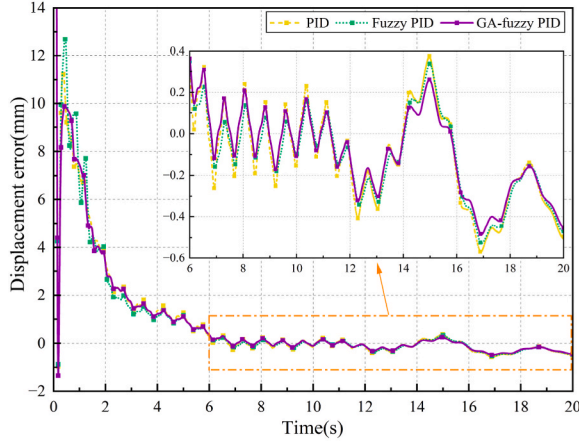
proposed controller also has better motion stability, and can enable quick adjustment, especially in the transition period of the motion. The suppression of the driving force fluctuations caused by external loads and internal parameter disturbances is attributed to the optimized fuzzy rules. It is evident that the stability of the limbs using the original fuzzy PID controller is not as good as that of the optimized controller.

Table 10 compares the average displacement errors of the six limb actuators with three control algorithms. During the motion for sea state levels 3 to 5, the errors are maintained near a more stable value, and the average displacement error of the six actuators with our proposed strategy are reduced to 1.5529, 1.6937 and 1.9869 mm, respectively. An average error reduction rate of 20.90 % is achieved for obvious control

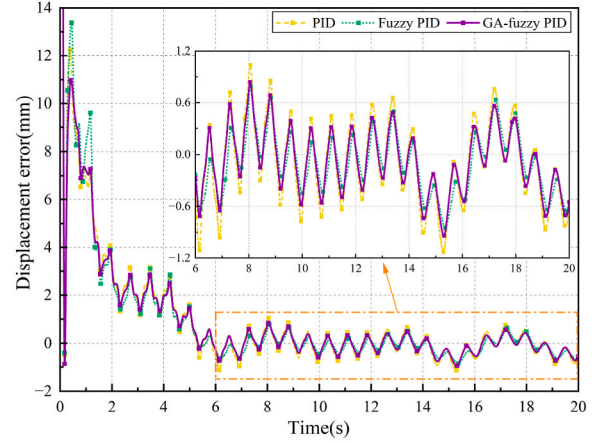
Table 9

Pose errors of the motion platform with different control strategies.

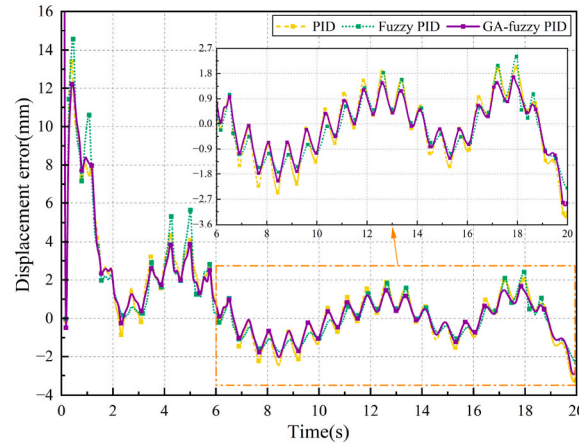
Sea state	Control strategy	X (mm)	Y (mm)	Z (mm)	RX (°)	RY (°)	RZ (°)
Level 3	PID	0.4859	0.0392	1.1058	0.0011	0.0166	0.0001
	Fuzzy PID	0.4835	0.0404	1.8006	0.0028	0.0189	0.0003
	GA-fuzzy PID	0.4797	0.0378	1.0799	0.0011	0.0152	0.0001
Level 4	PID	0.4298	0.0331	2.4939	0.0026	0.0428	0.0001
	Fuzzy PID	0.4246	0.0357	1.9691	0.0040	0.0423	0.0004
	GA-fuzzy PID	0.4236	0.0318	1.9416	0.0021	0.0353	0.0001
Level 5	PID	0.3540	0.0288	2.7416	0.0029	0.0637	0.0002
	Fuzzy PID	0.3314	0.0257	2.3312	0.0038	0.0552	0.0006
	GA-fuzzy PID	0.3313	0.0227	2.2672	0.0024	0.0487	0.0002



(a) Sea state level 3



(b) Sea state level 4



(c) Sea state level 5

Fig. 18. Displacement errors of limb 3 under three control strategies.**Table 10**

Displacement error comparison of limbs (mm).

Control strategy	Level 3	Level 4	Level 5
PID	2.0596	2.1519	2.3883
Fuzzy PID	1.5690	1.7181	2.0497
GA-fuzzy PID	1.5529	1.6937	1.9869

correction benefits, especially under the condition of high movement and heavy load.

To characterize the performance of the control strategies in a typical sea state level 5, the root mean square errors (RMSE) of position and attitude are used to describe the motion misalignment, while those of

the corresponding velocity are used to describe the motion stationarity, as shown in Eqs. (14) and (15) (Chen et al., 2023a).

$$\text{RMSE } p = \sqrt{\frac{\sum_{i=1}^n (p_i - \hat{p})^2}{n}} \quad (14)$$

$$\text{RMSE } v = \sqrt{\frac{\sum_{i=1}^n (\dot{p}_i - \hat{\dot{p}})^2}{n}} \quad (15)$$

where $\text{RMSE } p$ represents the RMSE of the position or attitude, $\text{RMSE } v$ represents the RMSE of the differential of position or attitude, p_i is the

Table 11

Motion deviation and stationarity in 6-DOF.

	Control strategy	X (mm)	Y (mm)	Z (mm)	RX (°)	RY (°)	RZ (°)
RMSE p	Fuzzy PID	0.3721	0.0448	9.580	0.0060	0.0677	0.0011
	GA-fuzzy PID	0.3719	0.0497	9.518	0.0043	0.0587	0.0005
RMSE v	Fuzzy PID	0.3738	0.1808	114.3	0.1016	0.2186	0.0132
	GA-fuzzy PID	0.2107	0.0994	114.0	0.0217	0.0823	0.0044

residual motion of the motion platform, $\hat{p}$ is the desired target value, and n is the number of data.

Table 11 shows the compensation deviation and stationarity in 6-DOF for both control strategies before and after optimization. The results show that the proposed control strategy can improve the position and attitude stationarity and motion stationarity. Although a smaller controller gain may result in larger residual motion, it contributes better stationarity, which makes the control strategy cope with the uncertainty factors under different sea conditions and satisfy the adaptability of different wave types.

From the comparison, the GA-fuzzy PID controller achieves motion control and has excellent ability to control motion and driving force in a complex environment. Furthermore, its control accuracy and stability are significantly improved compared to the controller designed by experience. Thus, GA can be used to develop a satisfactory fuzzy controller for the simulator, when a suitable dynamic model is established.

6. Conclusions

In this study, an integrated physical system model, CFD simulations in multiple sea states, and the GA-fuzzy PID controller for a vessel motion simulator were studied. It provides a cost-effective and accurate solution for testing shipboard equipment in a controlled laboratory environment. The main results of this paper are as follows.

- (1) Based on kinematic and dynamic characterization analyses, a physical system model was established integrally by an MBD model, an input and inverse kinematics system model, a sensor system model, and a control system model. This framework realizes precise replication of 6-DOF vessel motions with integrated models, input diversity, and motion control accuracy.
- (2) The CFD simulations for the input system model with mesh-free numerical methods such as the LBM, LES turbulence and the WALE viscosity models were used to predict the dynamic coupling between waves and the vessel. It enabled diverse CFD-driven inputs for the simulator, and the obtained motion trends were consistent with the actual motions.
- (3) A comparative analysis of the physical system model simulation and test with inputs from CFD results was conducted to verify the feasibility and accuracy. The results demonstrate that both the proposed physical system model and actual system exhibit excellent performance and meet the requirements.
- (4) The GA-fuzzy PID controller with optimized fuzzy rules by GA was proposed for the control system model. The simulation showed that it had better control performance in terms of the motion accuracy, response time, robustness and stationarity than those of the original fuzzy PID controller and PID controller. The average displacement error of the six limb actuators was reduced 20.90 %.

Future work should focus on the analysis of nonlinear factors such as dynamic load variation, static and dynamic friction, and platform transmission stiffness on the response characteristics for the simulator. We further plan a series of comprehensive model tests to validate the accuracy and reliability of the full-scale CFD calculations in the future. A more innovative control strategy and optimization algorithms will be

considered to refine the physical system model.

CRediT authorship contribution statement

Weibin Ma: Writing – review & editing, Writing – original draft, Methodology, Investigation. **Peng Wang:** Supervision, Project administration, Funding acquisition. **Huachao Dong:** Writing – review & editing, Validation, Funding acquisition. **Xinjing Wang:** Writing – review & editing, Funding acquisition, Conceptualization. **Hengzhi Fang:** Visualization, Software, Formal analysis. **Xianxian Fan:** Software, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Huachao Dong, Ph.D. reports financial support was provided by National Natural Science Foundation of China. Xinjing Wang, Ph.D. reports financial support was provided by National Natural Science Foundation of China. Xinjing Wang, Ph.D. reports financial support was provided by National Basic Scientific Research Program. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Bingul, Z., Karahan, O., 2022. Real-time trajectory tracking control of stewart platform using fractional order fuzzy PID controller optimized by particle swarm algorithm. *Ind. Robot* 49 (4), 708–725. <https://doi.org/10.1108/ir-07-2021-0157>.
- Cai, Y., Zheng, S., Liu, W., et al., 2021. Adaptive robust dual-loop control scheme of ship-mounted stewart platforms for wave compensation. *Mech. Mach. Theor.* 164, 104406. <https://doi.org/10.1016/j.mechmachtheory.2021.104406>.
- Chen, W., Wang, S., Li, J., et al., 2023a. An ADRC-based triple-loop control strategy of ship-mounted stewart platform for six-DOF wave compensation. *Mech. Mach. Theor.* 184, 105289. <https://doi.org/10.1016/j.mechmachtheory.2023.105289>.
- Chen, W., Wen, Y., Tong, X., et al., 2023b. Dynamics modeling and modal space control strategy of ship-borne stewart platform for wave compensation. *J. Mech. Robot.* 15 (4), 041015. <https://doi.org/10.1115/1.4062177>.
- Chen, Z., Zhan, G., Jiang, Z., et al., 2024. Adaptive impedance control for docking robot via stewart parallel mechanism. *ISA Trans.* 155, 361–372. <https://doi.org/10.1016/j.isatra.2024.09.008>.
- Chu, X., Han, W., Zhang, K., et al., 2016. Progress on simulating nonlinear free-surface flows with lattice boltzmann method. In: 2nd Conference of Global Chinese Scholars on Hydrodynamics (CCSH '2016), pp. 380–385.
- Dai, Y., Su, Q., Zhang, Y., 2020. A new dynamic model and trajectory tracking control strategy for deep ocean mining vehicle. *Ocean Eng.* 216, 108162. <https://doi.org/10.1016/j.oceaneng.2020.108162>.
- Dai, Y., Xue, C., Su, Q., 2021. An integrated dynamic model and optimized fuzzy controller for path tracking of deep-sea mining vehicle. *J. Mar. Sci. Eng.* 9 (3), 249. <https://doi.org/10.3390/jmse9030249>.
- Deep, K., Thakur, M., 2007. A new mutation operator for real coded genetic algorithms. *Appl. Math. Comput.* 193 (1), 211–230. <https://doi.org/10.1016/j.amc.2007.03.046>.
- Donatini, L., Verwilligen, J., Delefortrie, G., et al., 2025. Real-time synthesis of the dynamic responses of floating bodies to waves for maritime simulators. *Appl. Ocean Res.* 154, 104393. <https://doi.org/10.1016/j.apor.2024.104393>.

- Furqan, M., Suhaib, M., Ahmad, N., 2017. Studies on stewart platform manipulator: a review. *J. Mech. Sci. Technol.* 31 (9), 4459–4470. <https://doi.org/10.1007/s12206-017-0846-1>.
- Ganesan, T.S., Sen, D., Negi, A., 2024. Large amplitude ship motion simulation using a 3D numerical wave tank approach. *J. Ocean Eng. Mar. Energy* 10 (4), 879–916. <https://doi.org/10.1007/s40722-024-00356-8>.
- Gu, M., Song, B., 2017. Experimental trials of a sealed ocean kinetic energy harvester for underwater mooring platforms. *Adv. Mech. Eng.* 9 (10), 1687814017714971. <https://doi.org/10.1177/1687814017714971>.
- He, T., Feng, D., Liu, L., et al., 2022. CFD simulation and experimental study on coupled motion response of ship with tank in beam waves. *J. Mar. Sci. Eng.* 10 (1), 113. <https://doi.org/10.3390/jmse10010113>.
- Huang, S., Jiao, J., Chen, C., 2021. CFD prediction of ship seakeeping behavior in bi-directional cross wave compared with in uni-directional regular wave. *Appl. Ocean Res.* 107, 102426. <https://doi.org/10.1016/j.apor.2020.102426>.
- Jang, J., Bewley, T., 2023. Workspace analysis for parameter optimization of a cable-driven boat motion simulator. In: 2023 5th International Conference on Control and Robotics (ICCR), pp. 47–52. <https://doi.org/10.1109/ICCR60000.2023.10444880>.
- Kim, M., Hizir, O., Turan, O., et al., 2017. Numerical studies on added resistance and motions of KVLCC2 in head seas for various ship speeds. *Ocean Eng.* 140, 466–476. <https://doi.org/10.1016/j.oceaneng.2017.06.019>.
- Li, B.C., Lv, F., Li, X.F., 2012. Control system design and research of parallel ship motion simulator. *Adv. Mech. Eng.* 1810 (182–183), 1377–1382.
- Li, X., Shen, H., Yin, Y., 2021. Research on optimization of target ship system in ship handling simulator. In: 2021 5th International Conference on Vision, Image and Signal Processing (ICVISP), pp. 126–130. <https://doi.org/10.1109/ICVISP54630.2021.00030>.
- Liu, C., Wang, J., Wan, D., 2018. CFD computation of wave forces and motions of DTC ship in oblique waves. *Int. J. Offshore Polar Eng.* 28 (2), 154–163. <https://doi.org/10.17736/ijope.2018.sh21>.
- Liu, L., Chen, M., Yu, J., et al., 2022. Numerical study of swell impact on seakeeping performance of surface combatant. *Ocean Eng.* 249, 110827. <https://doi.org/10.1016/j.oceaneng.2022.110827>.
- Liu, T., Lou, H., Huang, H., 2023. Optimal design and verification of the 6-DOF stewart structure used in vehicle-mounted automatic assembly. *J. Phys. Conf. Ser.* 2591.
- Lu, Y., Wu, J., Li, W., et al., 2020. A new six-DOF parallel mechanism for captive model test. *Pol. Marit. Res.* 27 (3), 4–15. <https://doi.org/10.2478/pomr-2020-0041>.
- Min, D., Huang, D., Su, H., et al., 2019. High-precision tracking of cubic stewart platform using active disturbance rejection control. In: 38th Chinese Control Conference (CCC), pp. 3102–3107. <https://doi.org/10.23919/chicc.2019.8866606>.
- Ni, S., Liu, Z., Cai, Y., et al., 2020. A practical approach to numerically predicting a maneuvering vessel in waves oriented to maritime simulator. *Math. Probl Eng.* 2020, 8361951. <https://doi.org/10.1155/2020/8361951>.
- Ono, T., Eto, R., Yamakawa, J., et al., 2019. Control simulations for a stewart platform compensator mounted on moving base. *ASME Int. Mech. Eng. Congress Expo. (IMECE)* 4, 10780. <https://doi.org/10.1115/IMECE2019-10780>.
- Qiong, H., Bing-jie, L., Hao, Z., 2013. A 6-DOF deep-ocean mining ship motion simulator. *Inf. Technol. J.* 12, 8343–8348. <https://doi.org/10.3923/ijtj.2013.8343.8348>.
- Silva, D., Garrido, J., Riveiro, E., 2022. Stewart platform motion control automation with industrial resources to perform cycloidal and Oceanic wave trajectories. *Machines* 10 (8), 711. <https://doi.org/10.3390/machines10080711>.
- Stoker, J.J., 1957. *Water Waves: the Mathematical Theory with Applications*.
- Terrero-Gonzalez, A., Dai, S., Neilson, R.D., et al., 2024. Dynamic response of a shallow-draft floating wind turbine concept: experiments and modelling. *Renew. Energy* 226, 120454. <https://doi.org/10.1016/j.renene.2024.120454>.
- Wang, Z., Li, T., Ren, J., et al., 2022. URANS calculation of ship heave and pitch motions in marine simulator based on overset mesh. *J. Mar. Sci. Eng.* 10 (10), 1374. <https://doi.org/10.3390/jmse10101374>.
- Watanabe, S., Aoki, T., 2021. Large-scale flow simulations using lattice boltzmann method with AMR following free-surface on multiple GPUs. *Comput. Phys. Commun.* 264, 107871. <https://doi.org/10.1016/j.cpc.2021.107871>.
- Watanabe, S., Jun, K., Takayuki, A., et al., 2023. Free-surface flow simulations with floating objects using lattice boltzmann method. *Eng. Appl. Comp. Fluid Mech.* 17 (1), 2211143. <https://doi.org/10.1080/19942060.2023.2211143>.
- Wu, P., Chen, C., Zheng, X., et al., 2018. Comparative analysis on the methods of preliminarily estimating resistance of yachts. In: International Joint Conference on Materials Science and Mechanical Engineering (CMSME), vol. 383. <https://doi.org/10.1088/1757-899x/383/1/012052>.
- Xia, H., Wang, P., Jin, Z., et al., 2020. Maneuverability analysis of thrust vectoring ducted propeller with deflector. *Ocean Eng.* 213, 107614. <https://doi.org/10.1016/j.oceaneng.2020.107614>.
- Yin, L., Qiao, D., Li, B., et al., 2022. Modeling and controller design of an offshore wind service operation vessel with parallel active motion compensated gangway. *Ocean Eng.* 266, 112999. <https://doi.org/10.1016/j.oceaneng.2022.112999>.
- Zhang, Y., Guo, J., Peng, H., 2023. Simulation of offshore wind turbine blade docking based on the stewart platform. *Energy Eng* 120 (11), 2489–2502. <https://doi.org/10.32604/ee.2023.029496>.
- Zhao, Z., Liu, Y., Yu, C., et al., 2024. Fuzzy control-based adaptive adjustment of dynamic stiffness for stewart platforms. *Int. J. Adv. Comput. Sci. Appl.* 15 (6), 201–210.